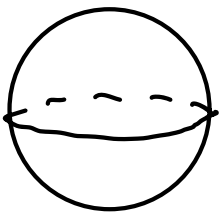
Lecture 3 // Sept 1 // Examples & Hodge Theory

Recall
- A connection is a map $\nabla : \Gamma(E) \to \Gamma(T^*M \otimes E)$
- Locally, connections take the form $\nabla = d + A$ where $A$ is a connection 1-form matrix
- Under change of coordinates, $A$ transforms strangely, not by conjugation: $A \mapsto g^{-1}Ag + g^{-1}dg$ where $g$ is bundle trans. fxn
- The Curvature of a connection is $F_A := dA + A \wedge A$ and by Chern-Weil theory, the chern classes are $\det\left(I - \frac{t}{2\pi} F_A\right)$

Today: We'll explicitly compute one connection & its curvature to illustrate them in a toy example. This example will also generalize to help us understand the Yang-Mills moduli space.

Example: Connections on $\mathbb{CP}^1 = S^2$

Recall that $\mathbb{CP}^1 = \{[z:w] \mid z,w \in \mathbb{C}^2\}$ can be covered with 2-charts, the $z$ chart with all points $[z:w]$ but $[0:1]$, and the $w$ chart. The transition function is $w = \frac{1}{z}$



Now, consider the $\mathbb{C}$-line bundle over $\mathbb{CP}^1$ defined by the bundle transition function $g(z) = z^n$. This is the degree-$n$ $\mathbb{C}$-line bundle where $\mathbb{C}$-fibers get glued by $z^n$, viewing this as a $1 \times 1$ matrix over $\mathbb{C}$.

Fact These are all the $\mathbb{C}$ line bundles over $\mathbb{CP}^1$

What do connections on the degree-$n$ line bundle look like?

To define a connection, we must write down the connection com- 1-form matrix $A$. Since the bundle is a line bundle, $A$ is a $1 \times 1$ matrix over $\mathbb{C}$, ie, a function $\mathbb{C} \to \mathbb{C}$.

Since $A$ is purely local, we must define it in local coordinates. We'll consider the $z$ chart first.

**Prop** $A := -n \dfrac{\bar{z}\,dz}{1+|z|^2}$ defines a connection on the degree $n$ $\mathbb{C}$-line bundle over $\mathbb{CP}^1$

**Pf** To check that $A$ actually defines a connection, we must do the following:

① look at $A$ in the other chart

② show that $A$ extends smoothly over the point at $\infty$.

Each step requires its own coordinate change:

① $A \rightsquigarrow g^{-1}Ag + g^{-1}dg$, the bundle change of coordinates

② $w = \frac{1}{z}$, the manifold change of coordinates

① Recall that the bundle transition function is $g(z) = z^n$.

Thus

$$g^{-1}Ag + g^{-1}dg = z^{-n}\left(-n\,\frac{\bar{z}dz}{1+|z|^2}\right)z^n + z^{-n}\,d(z^n)$$

$$= -n\,\frac{\bar{z}dz}{1+|z|^2} + \frac{nz^{n-1}}{z^n}$$

$$= -n\,\frac{\bar{z}dz}{1+|z|^2} + \frac{n}{z}$$

② We now apply the the manifold change-of-coordinates: $w = \frac{1}{z}$

$$dz = d\left(\tfrac{1}{w}\right) = \frac{-dw}{w^2}$$

So $-n\,\dfrac{\bar{z}dz}{1+|z|^2} + \dfrac{n}{z}$ becomes

$$\frac{-n\left(\frac{1}{w}\left(\frac{-1}{w^2}dw\right)\right)}{1+\left|\frac{1}{w}\right|^2} + nw\left(\frac{-1}{w^2}\right)dw = -n\,\frac{dw}{w^3\left(1+\left|\frac{1}{w}\right|^2\right)} - n\,\frac{dw}{w}$$

$$= n\,\frac{dw}{w(1+|w|^2)} - n\,\frac{dw}{w}$$

$$= n\left(\frac{1}{w(1+|w|^2)} - \frac{1}{w}\right)dw$$

$$= n\left(\frac{1}{w(1+|w|^2)} - \frac{(1+|w|^2)}{w(1+|w|^2)}\right)dw$$

$$= n\left(\frac{-|w|^2}{w(1+|w|^2)}\right)dw$$

$$= \frac{-n\,\overline{w}\,dw}{1+|w|^2}$$

this is a valid 1-form on the $w$-chart, and is defined at $z=\infty \leftrightarrow w=0$

Thus $A$ defines a valid global 1-form. □

Remark $A$ itself does not extend as a smooth 1-form on $\mathbb{CP}^1$:

$n\,\frac{dw}{w(1+|w|^2)}$ is singular at $w=0 \leftrightarrow z=\infty$, and it is the addition of the $-n\,\frac{dw}{w}$ term from the $g^{-1}dg$ in the bundle change of coordinates which resolves the issue, making $A$ globally defined.

So $A$ can be defined globally <u>because</u> of the "strange" bundle change-of-coordinates.

We'll now compute the curvature of this 1-form: $F_A = dA + A \wedge A$.

Since $d^2 = 0$, $A \wedge A = 0$, so we need only to compute $dA$.

$$dA = d\left(-n \frac{\bar{z}\, dz}{1+|z|}\right)$$

$$= -n \left( \frac{(1+|z|^2)\, d\bar{z} - \bar{z}\left(z\, d\bar{z} + \bar{z}\, dz\right)}{(1+|z|^2)^2} \right) \wedge dz$$

$$= -n \frac{d\hat{z} \wedge dz}{(1+|z|^2)^2} \qquad d\bar{z} = dx - i\,dy$$

$$dz = dx + i\,dy$$

$$= -2ni \frac{dx \wedge dy}{(1+x^2+y^2)^2}$$

Thus $F_A = -2ni \dfrac{dx \wedge dy}{(1+x^2+y^2)^2}$.

Computing the first chern class, we get that

$\Rightarrow$ first chern class: $\det\left(I - \frac{1}{2\pi i} F_A\right) = 1 + c_1$

$= $ Thus $c_1 = \frac{-1}{2\pi i}\left(-2ni \frac{dx \wedge dy}{(1+x^2+y^2)^2}\right) = \frac{n}{\pi} \frac{dx \wedge dy}{(1+x^2+y^2)^2}$

(note that this is the expression of $c_1$ on the $z$-chart)

observe: $$\int_{\mathbb{CP}^1} c_1 = n \left( \int_{\mathbb{R}^2} \frac{dx \wedge dy}{(1+x^2+y^2)^2} \right)$$

The previous computation leaves us with some questions.

1) If the final integral was taken over $\mathbb{R}^2$, where did $\mathbb{CP}^1$ come in to play? Why can't we choose another $A$ defined only on $\mathbb{R}^2$?

2) In this example,

$$\int_{\mathbb{CP}^1} c_1 = \int_{\mathbb{CP}^2} F_A = \int_{\mathbb{CP}^2} dA = \int_{\text{large disk}} dA = \int_{\text{large } S^1} A .$$

Could we instead choose $A$ to be compactly supported, so $\int = 0$? Or even choose $A = 0$?

The answer to both is a resounding no. The point here is that not every choice of $A$ defines a connection on a $U(1)$ bundle over $\mathbb{CP}^1$. In this example, even though $A$ is integrated over $\mathbb{R}^2$, we already checked that it extends, so the fact that $A$ is defined globally comes first.

$A$ "knows" what bundle it lives on by "remembering" the singular behavior at $z = \infty$, since it must match the $\frac{n\,dw}{w}$ term to allow for cancellation.

If $A \equiv 0$ or is compactly supported, $\dfrac{dw}{w(1+|w|^2)} = 0$,

so $$n\frac{dw}{w(1+|w|^2)} - n\frac{dw}{w} = -n\frac{dw}{w}.$$

For this to be nonsingular, $n$ must be $0$, so the line bundle must be trivial.

Upshot: The curvature integral determines the bundle via the behaviour near $\infty$.

We'll now review Hodge Theory:

Let $(M, g)$ be an oriented Riemannian manifold.

On each tangent space $T_pM$, we get an inner product pairing $\langle \cdot, \cdot \rangle_p$ which smoothly varies as $p$ varies in $M$.

This gives an inner product on each cotangent space $T_p^*M$, which induces an inner product on $\Lambda^k T^*M$ for each $k$ as follows:

$\{e_1, \dots, e_n\}$ oriented orthonormal basis of $T_pM$ $\leftrightarrow$ Oriented dual basis $\{e_1^*, \dots, e_n^*\}$ which we declare to be orthonormal. This declaration induces the inner product structure $\leftrightarrow$ for each $k$, we get $\binom{n}{k}$ basis vectors of the form $e_{i_1}^* \wedge \dots \wedge e_{i_k}^*$, where $\{i_1, \dots, i_k\} \subset \{1, \dots, n\}$. We declare this basis to be oriented and orthonormal, which induces the inner product structure

Of course, one must check that the resulting inner product doesn't depend on the original choice of oriented orthonormal basis for $T_pM$.

By abuse of notation, all of these inner products will be denoted by $\langle -, - \rangle$.

Rmk Let $n = \dim M$

On each $\Lambda^n T_p^* M$, there exists a unique oriented $n$-wedge with norm 1 via above identification.

This $n$-wedge is the volume form $e_1^* \wedge \dots \wedge e_n^* := \text{dvol}_g$

Now, note that since $\Lambda^k T^*M$ has $\binom{n}{k}$ basis vectors, $\dim(\Lambda^k T^*M) = \dim(\Lambda^{n-k} T^*M)$; this is also true for differential forms. To exploit this symmetry, we define the Hodge Star.

**Def** The Hodge Star is a map $* : \Omega^k \to \Omega^{n-k}$ defined pointwise by:

(after fixing an orthonormal oriented basis $\{e_1, \dots, e_n\}$ of $T_pM$)

For each $I \subset \{1, \dots, n\}$ with $|I| = k$, define $e_I^* := e_{i_1}^* \wedge \cdots \wedge e_{i_k}^*$

For each basis vector $e_I^*$, the hodge star sends $e_I^*$ to $*e_I^*$ via:

$$*e_I^* := \pm \underbrace{e_{j_1}^* \wedge e_{j_{n-k}}^*}_{e_J^*} \quad \text{where} \quad J = \{1, \dots, n\} \setminus I$$

The $\pm$ sign is chosen such that $e_I^* \wedge e_J^* = e_1^* \wedge \cdots \wedge e_n^*$,

so $*$ gives complementary wedge of forms.

To define $*$ on non-basis vectors, we extend linearly.

**Ex** $e_I = e_1 \wedge e_3$, $e_J = -e_2 \wedge e_4$

$\Rightarrow e_I \wedge e_J = e_1 \wedge e_3 \wedge -e_2 \wedge e_4$

$= e_1 \wedge e_2 \wedge e_3 \wedge e_4 = dvol$

**Rmk** 1) This definition again is independent of the choice of orthonormal basis, but it <u>does</u> depend on the choice of metric.

2) $*$ is <u>almost</u> an involution... but not quite:

$$(*)^2 = (-1)^{k(n-k)}$$

$e_I^* \mapsto e_J^*$ s.t. $e_I^* \wedge e_J^* = dvol$

$e_J^* \mapsto e_I^*$ s.t. $e_I^* \wedge e_J^* = dvol$

} total # swaps depends on $k$.

Defining the Hodge star immediately buys us two things.

First, we can get an inner product structure on $\Omega^k$ from that on $\Lambda^k T^*M$, which was induced by the metric.

**Def** For $\alpha, \beta \in \Omega^k$, define

$$\langle \alpha, \beta \rangle := \int_M \alpha \wedge * \beta \quad \text{since } \alpha \wedge * \beta = n\text{-form.}$$

If $\alpha$ is an orthonormal basis element, $\alpha \wedge * \alpha = d\text{vol}_g$, so

$$\langle \alpha, \alpha \rangle = \int_M |\alpha|^2 d\text{vol}_g$$

Second, we get a formal adjoint of the exterior derivative.

**Recall** The exterior derivative $d : \Omega^k \to \Omega^{k+1}$

**Def** The "codifferential" $d^* : \Omega^k \to \Omega^{k-1}$ is defined as the composition $d^* := (-1)^k * d *$

$$d^* : \Omega^k \xrightarrow{*} \Omega^{n-k} \xrightarrow{d} \Omega^{n-k+1} \xrightarrow{*} \Omega^{k-1}$$

**Lemma** $d^*$ is the formal adjoint of $d$: $\langle d\alpha, \beta \rangle = \langle \alpha, d^* \beta \rangle$

**Pf** By Stokes' theorem and the product rule, $*(*d*)\beta$ $\overbrace{\quad\quad}^{\pm 1}$

$$0 = \int_M d(\alpha \wedge * \beta) = \int_M d\alpha \wedge * \beta \pm \int_M \alpha \wedge \underbrace{d(* \beta)}$$

$$= \langle d\alpha, \beta \rangle \pm \int_M \alpha \wedge * d^* \beta$$

$$= \langle d\alpha, \beta \rangle \pm \langle \alpha, d^* \beta \rangle$$

$$\Rightarrow \langle d\alpha, \beta \rangle = \mp \langle \alpha, d^* \beta \rangle \quad \square$$

Note $(d^*)^2 = 0$

We now have the tools to discuss the main point of Hodge theory (for our purposes).

Def The Hodge Laplacian is a map $\Delta : \Omega^k \to \Omega^k$ defined as

$$\Delta = (d + d^*)^2 = dd^* + d^*d$$

Def $\alpha \in \Omega^k$ is called harmonic if $\Delta\alpha = 0$

Rmk In local coordinates, $d^*d + dd^*$ is a 2nd order differential operator, since it acts like a $2^{nd}$ derivative on the coefficients of basis vectors.

Lemma $\alpha$ is harmonic IFF $d\alpha = 0$ (closed) and $d^*\alpha = 0$ (co-closed)

Pf ($\Leftarrow$) obvio

($\Rightarrow$) If $\Delta = 0$, $\langle \Delta\alpha, \alpha \rangle = 0$. Since $\Delta\alpha = dd^*\alpha + d^*d\alpha$,

$$\begin{aligned} 0 = \langle \Delta\alpha, \alpha \rangle &= \langle dd^*\alpha + d^*d\alpha, \alpha \rangle \\ &= \langle dd^*\alpha, \alpha \rangle + \langle d^*d\alpha, \alpha \rangle \\ &= \langle d^*\alpha, d^*\alpha \rangle + \langle d\alpha, d\alpha \rangle \\ &= \langle (d + d^*)\alpha, (d + d^*)\alpha \rangle \end{aligned}$$

$$\Rightarrow (d + d^*)\alpha = 0$$

$$\Rightarrow d\alpha = d^*\alpha = 0 \quad \square$$

Note If $\alpha$ is harmonic, it is closed, so $\alpha$ defines a cohomology class $[\alpha] \in H^*_{dR}$

Lemma $\alpha$ is harmonic IFF $\alpha$ minimizes norm in its de Rham cohomology class

Pf ($\Rightarrow$) Let $\alpha \in H^k_{dR}(M;\mathbb{R})$.

Every other element in this class will differ from $\alpha$ by an exact form, ie, some $d\beta$ for some $\beta \in \Omega^{k-1}$. We will compute the norm of $\alpha + d\beta$.

$$\langle \alpha + d\beta, \alpha + d\beta \rangle = \langle \alpha, \alpha \rangle + \langle d\beta, d\beta \rangle + 2\langle \alpha, d\beta \rangle$$

We know that $\langle d\beta, d\beta \rangle \geq 0$, and $\langle \alpha, d\beta \rangle = \langle d^*\alpha, \beta \rangle = 0$ since $\alpha$ is harmonic.

Thus $\langle \alpha, \alpha \rangle \leq \langle \alpha + d\beta, \alpha + d\beta \rangle$. □

($\Leftarrow$) Assume $\alpha$ is norm minimizing.

For all $\beta \in \Omega^{k-1}$, consider the ray $\langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle_{t \in [0,1]}$.

Eg, we move $\alpha$ around in its cohomology class along some ray.

Since $\alpha$ is norm minimizing, when $t=0$, the derivative vanishes.

$$0 = \frac{d}{dt}\Big|_{t=0} \langle \alpha + t\,d\beta, \alpha + t\,d\beta \rangle = 2\langle \alpha, d\beta \rangle$$

$$= 2\langle d^*\alpha, \beta \rangle$$

$$\Rightarrow d^*\alpha = 0$$

It remains to show $d\alpha = 0$ and this minimizer is unique; the proof is left as an exercise. □

Thm Hodge Theorem

Define $\mathcal{H}^k$ to be the space of harmonic $k$-forms.

There exists an isomorphism $\mathcal{H}^k \cong H^k_{dR}$ sending $\alpha \in H^k_{dR}$ to the norm minimizer in $[\alpha]$.